

# News Facilitator Intent Validation: Comparative Performance Report

This report summarizes the experimental results across four distinct model runs on March 31, 2026. The goal was to validate the **Supervisor Agent's** ability to decompose user prompts into four intents: **Business News**, **Sports News**, **General News**, and **Web Search**.

## 1. Executive Summary: The Performance Matrix

| Metric               | Groq: gpt-oss-120b | OpenAI: gpt-3.5-turbo | OpenAI: gpt-4o-mini | Groq: gpt-oss-20b |
|----------------------|--------------------|-----------------------|---------------------|-------------------|
| Exact Match Accuracy | 0.9 (90%)          | 0.9 (90%)             | 0.9 (90%)           | 0.8 (80%)         |
| Elapsed Time         | 7.82s              | 7.56s                 | 27.45s              | 3.33s             |
| Avg. Intent F1       | 0.9                | 0.9                   | 0.9                 | 0.8               |
| Provider             | Groq               | OpenAI                | OpenAI              | Groq              |

## 2. Key Findings & Model Analysis

### The "Gold Standard" Group (90% Accuracy)

Three models—**gpt-oss-120b**, **gpt-3.5-turbo**, and **gpt-4o-mini**—all achieved an impressive **90% accuracy**.

- **Best All-Rounder: gpt-3.5-turbo** delivered the top accuracy with the fastest performance in its tier (**7.56s**).
- **The Latency Warning: gpt-4o-mini** matched the intelligence of the others but was nearly **4x slower** than the other top-tier models, taking **27.45s** to complete the 10 prompts.

- **The Consensus Error:** Every single model (including the 90% group) failed on the "**Global Chip Shortage**" prompt. All models predicted **General News** while the ground truth expected **Business News**.

### The Efficiency Specialist (80% Accuracy)

- **Groq gpt-oss-20b:** While it had the lowest accuracy at **80%**, it was the undisputed speed champion, finishing the run in just **3.33 seconds**.
- **Failure Mode:** It was the only model to return an empty prediction for the complex "triple-intent" query (UEFA/USD-EUR/Mars Rover), suggesting it may struggle with high instruction density.

## 3. Per-Intent Performance Deep Dive

### Business News (The "Fuzzy" Boundary)

- All models demonstrated high precision but struggled with recall on topics like "Chip Shortages".
- **gpt-oss-20b** was the most conservative, with only **60% recall** in this category.

### Sports & Web Search (The "Locked" Categories)

- **Sports News:** All 90% accuracy models achieved **100% Recall and Precision**.
- **Web Search:** The **gpt-oss-120b**, **gpt-3.5-turbo**, and **4o-mini** models were flawless here (**100% across all metrics**).

## 4. Recommendations

1. **For Production Speed:** Use **Groq gpt-oss-20b**. Its 3.3s latency is ideal for real-time routing, provided you don't expect highly complex, multi-intent user queries.
2. **For Reliability:** Use **gpt-3.5-turbo**. It provides the highest accuracy with a very stable 7.5s response time.

3. **Prompt Adjustment:** Since all models disagreed with the ground truth on "Chip Shortages," consider moving supply-chain related topics into the **General News** category or providing a few-shot example to anchor them in **Business News**.